

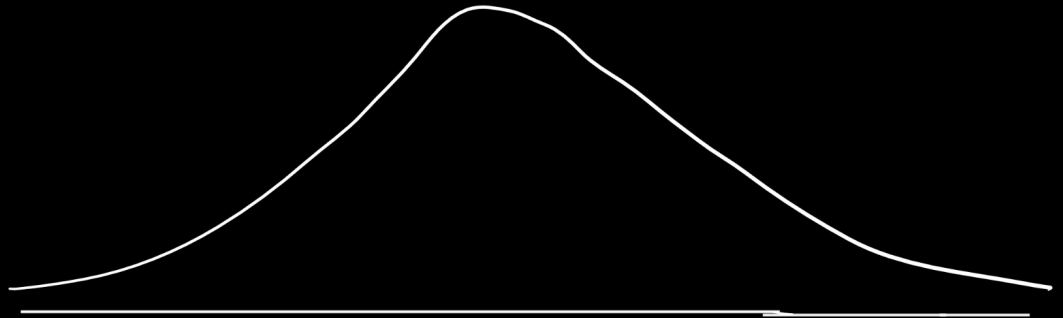
Random Variable

Descriptive
stats
 μ, σ

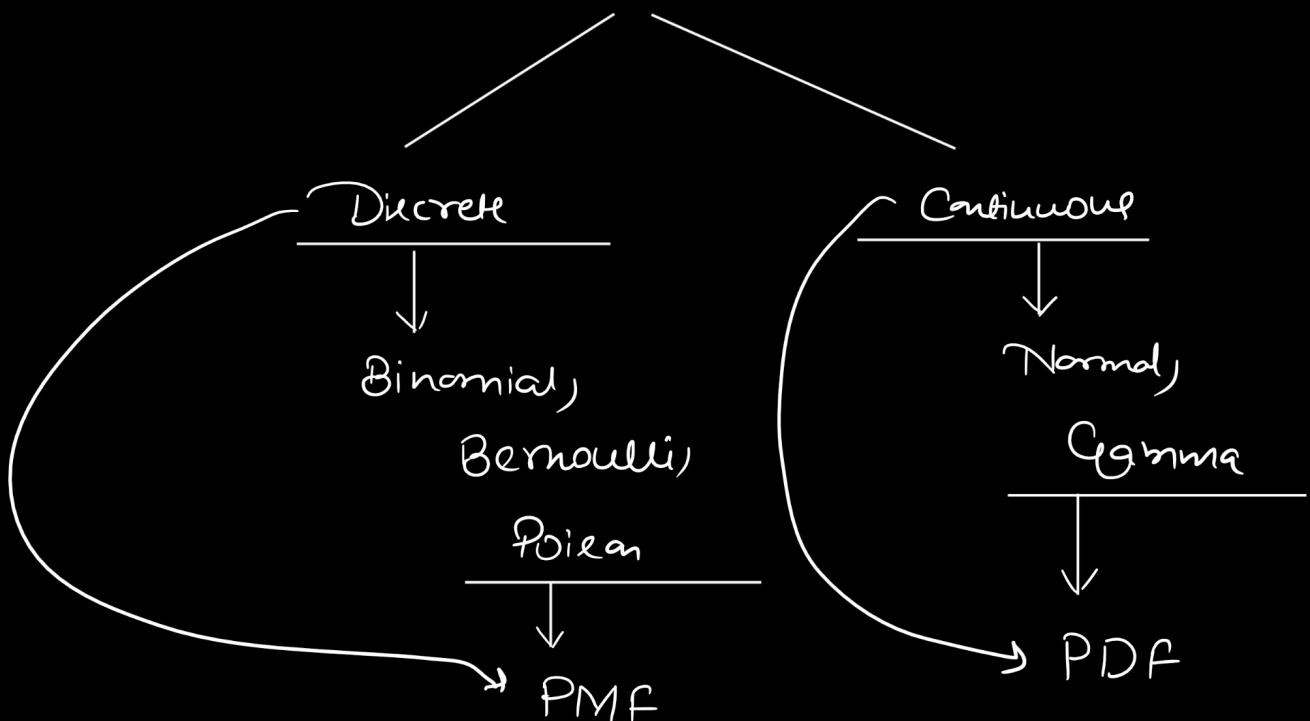
Jan $\rightarrow$ 80000
Feb $\rightarrow$ 150000
March $\rightarrow$ 85000
April $\rightarrow$??

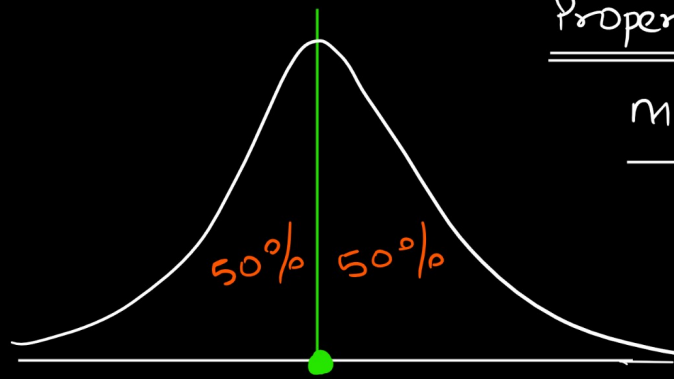
Probabilistic Modeling

Prediction for the April month



Random Variable





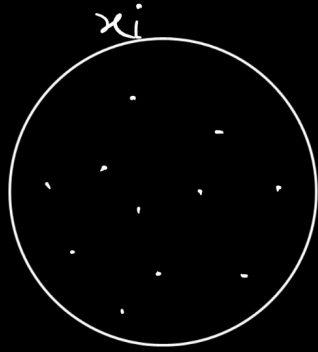
Properties

Mean = Median = Mode

Symmetrical

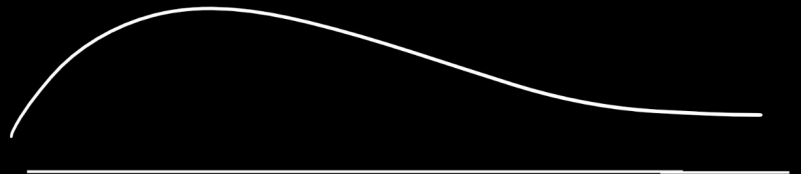
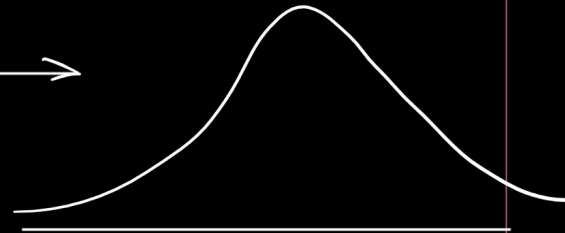
Distribution

Mode = Mean = Median



$\xrightarrow{PDF}$
 $f(x_i)$

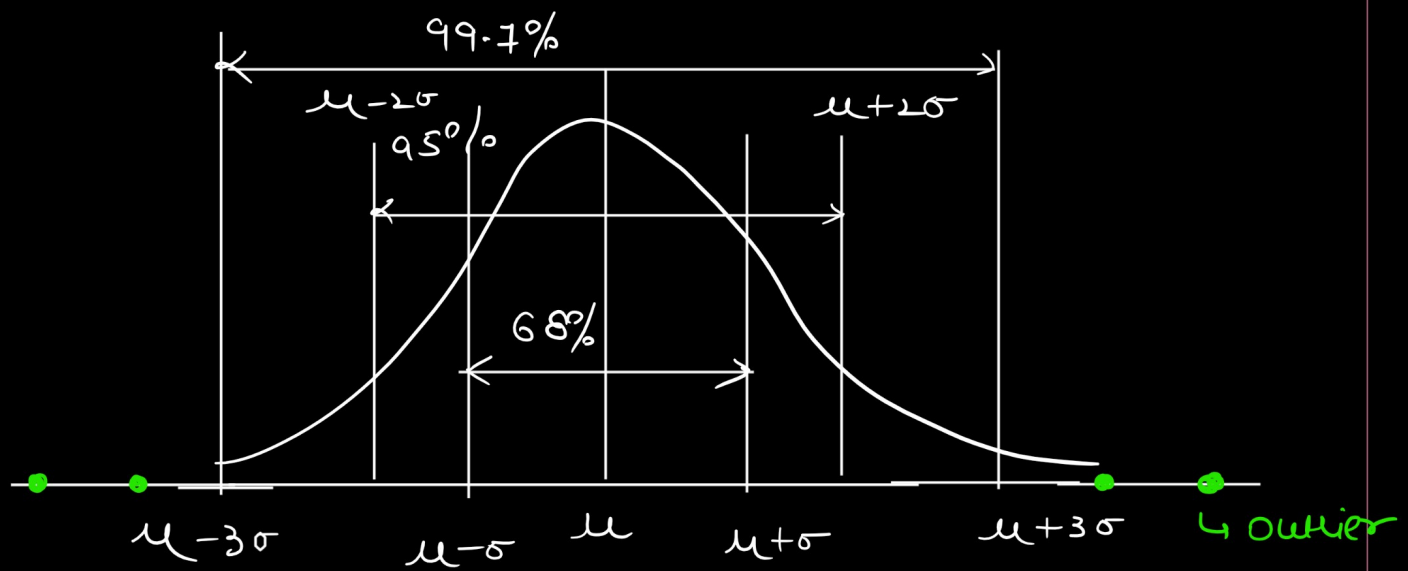
$\xrightarrow{PDF}$
 $f(x_i)$



<u>Age</u>		<u>SNF</u>
44	—	<u>2.79</u>
47	—	<u>3.42</u>
65	—	<u>4.69</u>
62	—	<u>1.26</u>
73	—	<u>2.43</u>

Standardize the values
Within a given
specific
range

$\mu = 0, \sigma = 1$



$\left. \begin{array}{l} \mu = 70 \\ \sigma = 10 \end{array} \right\} \rightarrow 68\% \text{ of}$
Students scored
b/w what

$\mu - \sigma \text{ to } \mu + \sigma$
60 to 80

specific
range?

95% of
Students scored

b/w what
specific range

$\rightarrow \mu - 2\sigma$
 to
 $\mu + 2\sigma$

$70 - 2 \times 10 \text{ to } 70 + 2 \times 10$
50 to 90

99.7% of students score

b/w what range

↳ $70 - 3 \times 10$ to

$70 + 3 \times 10$

40 to 100

↖ 21 - 59

Age

↗ 21000 to 150000

Salary

↳ Standardize/Normalize

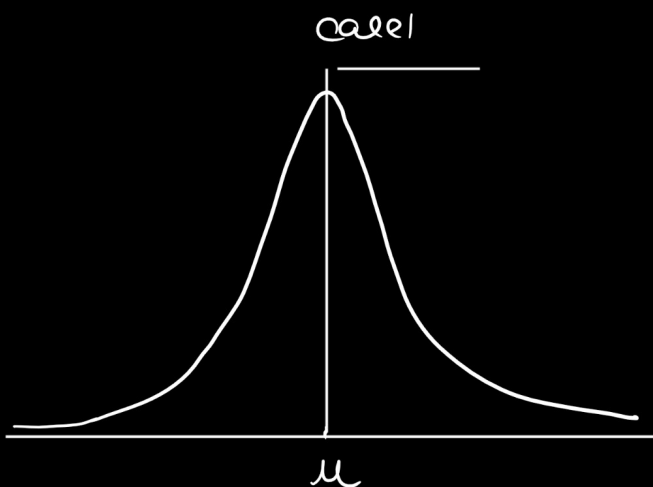
Within one

specific

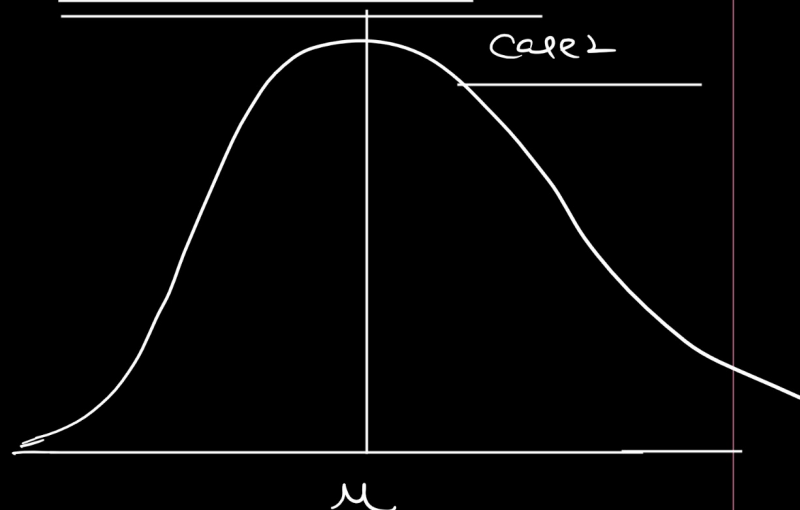
range

SNR →
$$Z = \frac{x_i - \mu}{\sigma}$$

$\mu = 0, \sigma = 1$



Low Variance



high Variance

Variance $\rightarrow$ uncertainty $\rightarrow$ How uncertain
the outcome are?
high variance $\rightarrow$ unstable, unreliable model

Dataset

Machine Learning

① training set

$\downarrow$ bias $\rightarrow$ high training error

② testing
set

$\downarrow$ Variance $\rightarrow$ high variance

$\hookrightarrow$ high testing error &
low training
error

$\times$ underfitting $\rightarrow$ high bias & low variance

$\times$ overfitting $\rightarrow$ low bias & high variance

Model Type	Bias	Variance
Very simple	High	Low
Very complex	Low	High



Value of r	Interpretation
+1	Perfect positive linear relationship
0.7 to 0.9	<u>Strong positive relationship</u>
0.3 to 0.7	Moderate positive relationship
0 to 0.3	Weak positive relationship
0	No linear relationship
-0.3 to -0.7	Moderate negative relationship
-0.7 to -1	<u>Strong negative relationship</u>

Conditional Probability

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

Bayes Theorem

$$P(A \cap B) = \underbrace{P(A | B)}_{\text{Likelihood}} * P(B)$$

$$P(B | A) = \frac{P(A \cap B)}{P(A)} = \frac{P(A | B) * P(B)}{P(A)}$$

Posterior Evidence Prior

Imagine you have three boxes:

Box 1 contains 2 red balls and 1 blue ball.

Box 2 contains 1 red ball and 2 blue balls.

Box 3 contains 3 red balls and 1 blue ball.

If you randomly choose a box and then randomly draw a ball from the chosen box, the probability of drawing a red ball can be calculated using the law of total probability.

law of total probability

$$P(B_1) = \frac{1}{3}$$

$$P(R|B_1) = \frac{2}{3}$$

$$P(B_2) = \frac{1}{3}$$

$$P(R|B_2) = \frac{1}{3}$$

$$P(B_3) = \frac{1}{3}$$

$$P(R|B_3) = \frac{3}{4}$$

$$P(R) = P(B_1) * P(R|B_1) + P(B_2) * P(R|B_2) + P(B_3) * P(R|B_3)$$

$$\frac{1}{3} * \frac{2}{3} + \frac{1}{3} * \frac{1}{3} + \frac{1}{3} * \frac{3}{4}$$

$$\approx \frac{7}{12}$$

Naive Bayes Theorem

↳ classification task

$$P(Y | \text{Patient feature}) \longrightarrow \underline{0.7}$$

$$\underline{P(N | \text{Patient feature})} \longrightarrow \underline{0.3}$$

- ↳ medical diagnosis
- ↳ fraud detection
- ↳ spam filtering